



मोतीलाल नेहरू राष्ट्रीय प्रौद्योगिकी संस्थान इलाहाबाद
प्रयागराज 211004 भारत
Motilal Nehru National Institute of Technology Allahabad
Prayagraj - 211004 (India)

School of Management Studies
Mid Semester (Even) Examination 2023-24

Semester: IVth

Programme Name: B.Tech.

Course Code: EAN14705

Course Name: Foreign Language (FRENCH)

Branch: All Branches

Student Reg.No.:

2 0 2 2 ~~2 0 2 2~~

Duration: 90 Minutes

Max. Marks: 25

[Marks]
[5.0]

Answer all questions:

Q1. Days of the week: Match the two columns.

A	*B*
Vendredi	Sunday
Mercredi	Tuesday
Dimanche	Wednesday
Mardi	Friday

[10.0]

Q 2. Translate any FIVE greetings in French

1. See you soon!
2. Have a good day!
3. Thank you very much.
4. Good evening!
5. See you on Saturday!
6. How are you?
7. Please.

Q3. Numbers: Write the numbers in French or in digits (5 marks)*

1. 27 -
2. Douze -
3. 61 -
4. 43 -
5. Cinquante-cinq

[5.0]

Q 4. Write the missing months (5 marks)

janvier - - mars - - mai - - - - septembre - octobre - novembre -
décembre.

[5.0]

Department of Computer Science & Engineering
Mid Semester Examination, Session 2024-25 (Even)

Semester: IV

Programme: B.Tech.
Course Name: Compiler Construction
Course Code: CSN 14402
Time: 1.5 Hours

Branch: CSE
Max. Marks: 25
Registration No.: 2022

Instructions:

1. Attempt any five questions. Each question carries 5 Marks each.
2. All parts of a question should be answered in one attempt SEQUENTIALLY.

2. Attempt any five questions. Each question carries 5 Marks each.			Marks																												
2. All parts of a question should be answered in one attempt SEQUENTIALLY.			5 [CO1]																												
Q1	a. What do you mean by Compiler? b. Explain the classification of parsing techniques . c. What do you mean by Lexical Analyzer? d. Explain the term Token, Pattern, Lexemes.																														
Q2	a. Determine the grammar generating the language $L = \{a^i b^{2i}\}$ b. Using that grammar find out the length of the derivation (number of steps starting from S) to generate the string $a^i b^{2i}$. Draw the respective automata for the above language.		5 [CO3]																												
Q3	a. Check whether the following grammar (with rules) is deterministic or not. If they are non-deterministic, convert it into an equivalent deterministic grammar. $S \rightarrow iEtS iEtSeS a$ $E \rightarrow b$ b. Explain the difference between top-down and bottom-up parsers with example.		5 [CO2]																												
Q4	Consider the following context-free grammar where the set of terminals is $S \rightarrow daT Rf$ $T \rightarrow aS baT \epsilon$ $R \rightarrow caTR \epsilon$ The following is partially filled LL(1) parsing table <table><tr><th></th><th>a</th><th>b</th><th>c</th><th>d</th><th>f</th><th>\$</th></tr><tr><td>S</td><td></td><td></td><td>1</td><td>$S \rightarrow daT$</td><td>2</td><td></td></tr><tr><td>T</td><td>$T \rightarrow aS$</td><td>$T \rightarrow baT$</td><td>3</td><td></td><td>$T \rightarrow \epsilon$</td><td>4</td></tr><tr><td>R</td><td></td><td></td><td>$R \rightarrow caTR$</td><td></td><td>$R \rightarrow \epsilon$</td><td></td></tr></table> a. Write the FIRST and FOLLOW sets for the non-terminals S,T, and R. b. Write the appropriate entry for 1,2,3, and 4.		a	b	c	d	f	\$	S			1	$S \rightarrow daT$	2		T	$T \rightarrow aS$	$T \rightarrow baT$	3		$T \rightarrow \epsilon$	4	R			$R \rightarrow caTR$		$R \rightarrow \epsilon$			5 [CO3]
	a	b	c	d	f	\$																									
S			1	$S \rightarrow daT$	2																										
T	$T \rightarrow aS$	$T \rightarrow baT$	3		$T \rightarrow \epsilon$	4																									
R			$R \rightarrow caTR$		$R \rightarrow \epsilon$																										



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Mid Semester (Even) Examination, 2023-24

Programme Name: B.Tech

Course Code: CSN14401

Branch: CSE

Duration: 1 Hour and 30 Minutes

Semester: IV

Course Name: Operating System

Student Reg. No.:

2	0	2	2						
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Max. Marks: 20

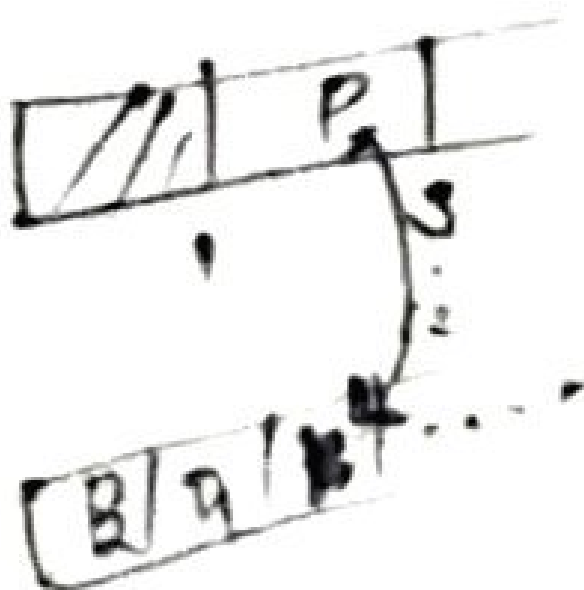
Instructions: (1) All questions are compulsory.
(2) Figures to the right indicate the full marks.

- Q 1. Illuminate the reasons why Linus Torvalds, the creator of Linux chosen monolithic kernel over microkernel as an architectural choice for Linux. Also, explain how Linux's monolithic kernel supports loadable kernel modules. [4]
- Q 2. How many child processes will be created in the following C program when it is executed on a Linux system? Explain your solution using parent-child process tree. [2]
- ```
#include<unistd.h>
int main() {
 int i = 0;
 for(++i; i<5; i++)
 if(i%2==0) fork(); fork();
 return 0; }
```
- Q 3. Explain the steps involved in the program's execution that lead to the creation of zombie and orphan processes. Further, describe how the operating system manages these processes, and what are the potential consequences of having an excessive number of zombie and orphan processes on a Unix/Linux system. [4]
- Q 4. Discuss how Inter-Process Communication (IPC) mechanisms can be employed to solve the producer-consumer problem in a multi-process cooperative environment. Provide a detailed explanation of how shared memory and message-passing techniques can facilitate communication between producer and consumer processes. Additionally, analyze the advantages and potential drawbacks of each IPC method in the context of addressing synchronization issues in the producer-consumer scenario. [4]
- Q 5. Consider the following set of processes, with the given arrival and burst times (in milliseconds): [6]

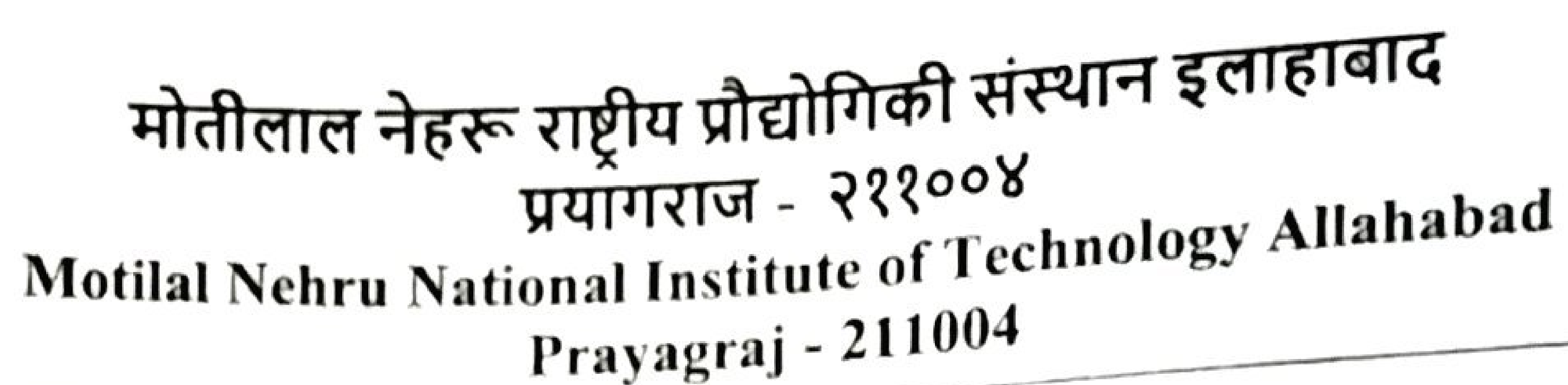
| Process        | Arrival Time | Burst Time |
|----------------|--------------|------------|
| P <sub>1</sub> | 3            | 4          |
| P <sub>2</sub> | 2            | 3          |
| P <sub>3</sub> | 1            | 5          |
| P <sub>4</sub> | 5            | 6          |
| P <sub>5</sub> | 4            | 2          |

- (a) Draw the Gantt charts illustrating the execution of these processes using non-preemptive LJF (Longest Job First) and Round Robin (quantum = 2) scheduling.
- (b) What is the average turnaround time for the scheduling algorithms in part (a)?
- (c) What is the average waiting time for the scheduling algorithms in part (a)?

Round - Robin







**Department of Computer Science & Engineering**  
**Mid Semester Examination, Session 2023-24 (Even)**

Programme: B.Tech.

**Course Name: Object Oriented Modelling**  
**Time: 90 Minutes**

**Branch: CSE**  
**Max. Marks: 25**

**Semester: IV**

**Semester: IV**  
**Course Code: CSN14101**

**Reg. No.**

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**Instructions:**

1. *This question paper contains 5 questions.*
2. *Attempt all the questions.*
3. *Each of the following question carries 5 marks.*

**Q1.**

[5]

Prepare a class diagram for a graphical document editor that supports grouping. Assume that a document consists of several sheets. Each sheet contains drawing objects, including text, geometrical objects, and groups. A group is simply a set of drawing objects, possibly including other groups. A group must contain at least two drawing objects. A drawing object can be a direct member of at most one group. Geometrical objects include circles, ellipses, rectangles, lines, and squares. While representing the classes and their relationships in the class diagram, ignore the attributes and methods associated with the classes.

**Q2.**

[5]

[5]

Categorize the following relationships into Generalization, Composition, Association, or Dependency. Beware, there may be N-ary Associations in the following list, so do not assume that every relationship involving three or more classes is a Generalization. Explain your answers with diagrammatical representation of Generalization, Composition, Association, or Dependency among different entities of the following relationships:

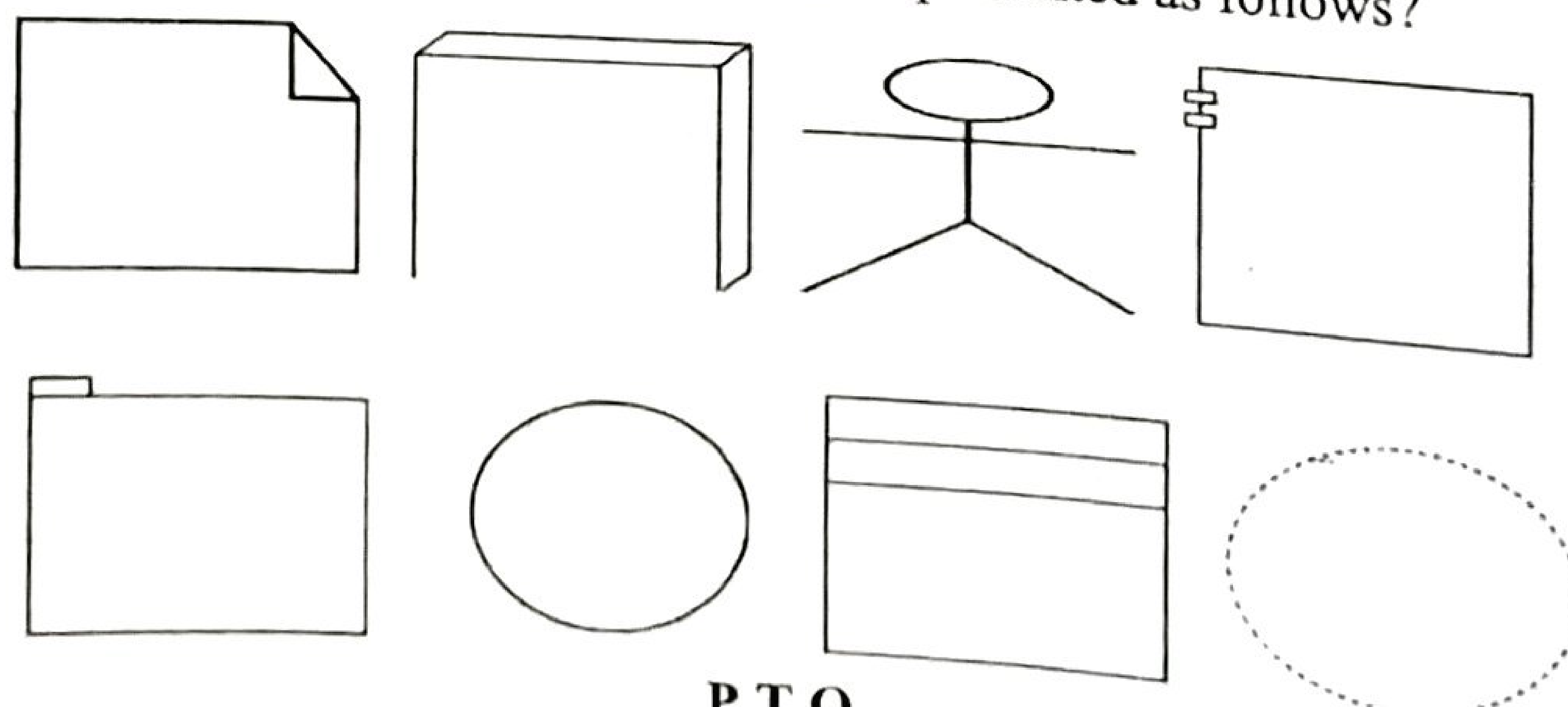
a) A dining table has four legs.

- a) A dining philosopher uses a fork.
- b) A file is an ordinary file or a directory file.
- c) File contains records.
- d) A drawing object is text, a geometrical object, or a group.
- e) A student is taught by a professor.

**Q3.**

**[1+2+2]**

- (a) Software developers, project managers, clients, and end-users perceive Unified Modelling Language (UML) in different views. What are the different views in UML? [1+2+2]
- (b) Things are considered as one of the building block in UML. Following notations are representing Structural, Grouping or Annotational things of UML. Identify which of the Structural, Grouping or Annotational things can be represented as follows?



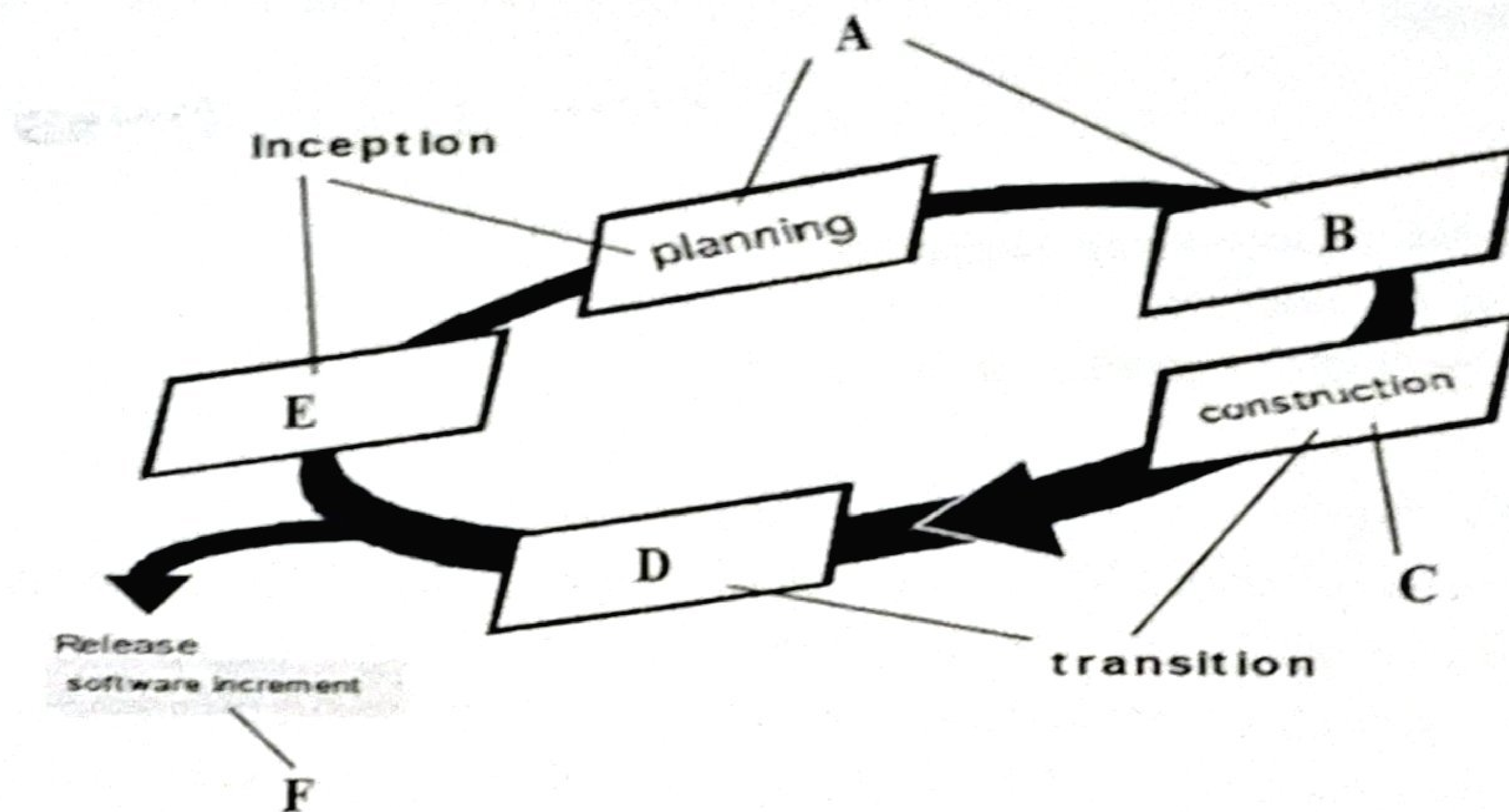
P.T.O.



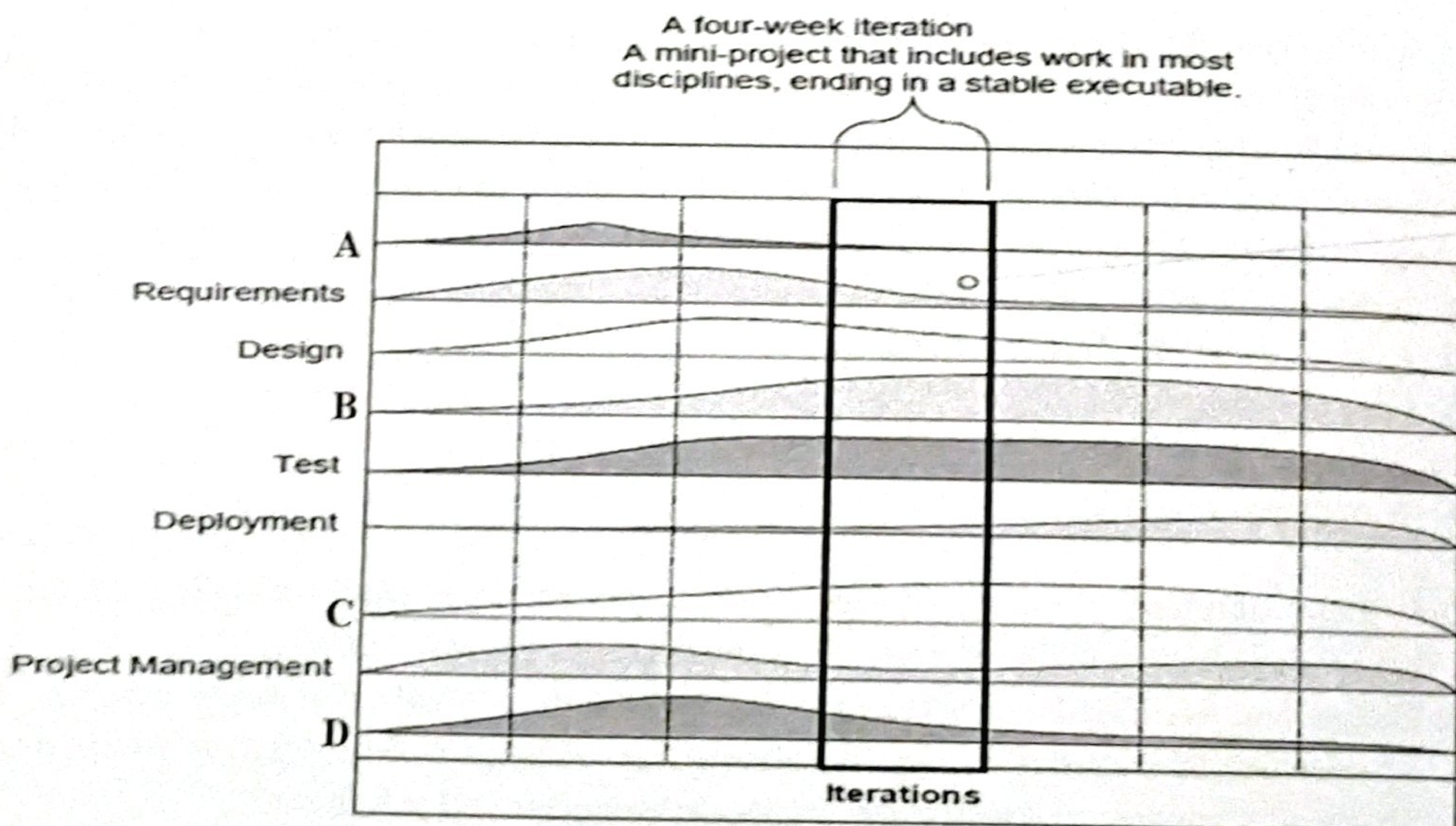
- (c) What is the purpose of utilizing Structural, Behavioural, and Interaction diagrams in UML? Mention the different UML diagrams which lie in the Structural, Behavioural, and Interaction categories. [3+2]

Q4.

- a) In the following diagram, different phases of unified process are shown in the form of iterative steps. Some of the entries are missing in the following diagram which are denoted as A, B, C, D, E, and F. Mention the activities represented by the missing entries.



- b) Following diagram represents several disciplines related to an iterative unified process for a mini-project development. Find out the missing entries denoted as A, B, C, and D.



Q5.

- a) Discuss the purpose of object modelling technique proposed by James Rumbaugh.  
b) Elaborate the different models associated with the object modelling technique suggested by James Rumbaugh.  
c) What are the sequential phases in object modelling technique developed by Grady Booch. Mention the iterative sub-steps involved in each phase of the object modelling technique proposed by Grady Booch. [1+2+2]





Department of Computer Science & Engineering  
Mid Semester Examination Session 2023-24 (Even)

|              |                         |                   |      |           |    |
|--------------|-------------------------|-------------------|------|-----------|----|
| Programme:   | B.Tech.                 | Branch:           | CSE  | Semester: | IV |
| Course Name: | Artificial Intelligence |                   |      |           |    |
| Course Code: | < CSN14403 >            |                   |      |           |    |
| Time:        | 90 Minutes              | Max. Marks:       | 20   |           |    |
|              |                         | Registration No.: | 2022 |           |    |

Instructions (related to question paper):

1. All questions are compulsory
2. Attempt the questions strictly in sequential order.
3. Calculator is allowed
4. Write assumptions correctly (in case you feel data is missing)

| Q1 | Write the answer of the following question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|    | <p>Three of your Players, Alvin, Sergey, and James rent a cookie shuffler, which takes in a set number of cookies and groups them into <b>3 batches</b>, one for each player. The cookie shuffler has three levers (with positions either UP or DOWN), which act to control how the cookies are distributed among the three players. Assume that <b>30 cookies</b> are initially put into the shuffler. Each player controls one lever, and they act in turn. Alvin goes first, followed by Sergey, and finally James. Assume that all players are able to calculate the payoffs for every player at the terminal nodes. Assume the payoffs at the leaves correspond to the number of cookies for each player in their corresponding turn order. Hence, a utility of <b>(7,10,13)</b> corresponds to Alvin getting 7 cookies, Sergey getting 10 cookies, and James getting 13 cookies. No cookies are lost in the process, so the sum of cookies of all three players must equal the number of cookies put into the shuffler. Players want to maximize their own number of cookies. (Refer Figure 1)</p> <p>a. What is the utility triple propagated up to the root?</p> <p>b. Is pruning possible in this game? "Yes" or "No". If Yes, cross out all nodes (both leaves and intermediate nodes) that get pruned. If no, explain in one sentence why pruning is not possible. Assume the tree traversal goes from left to right.</p> | [5]   |

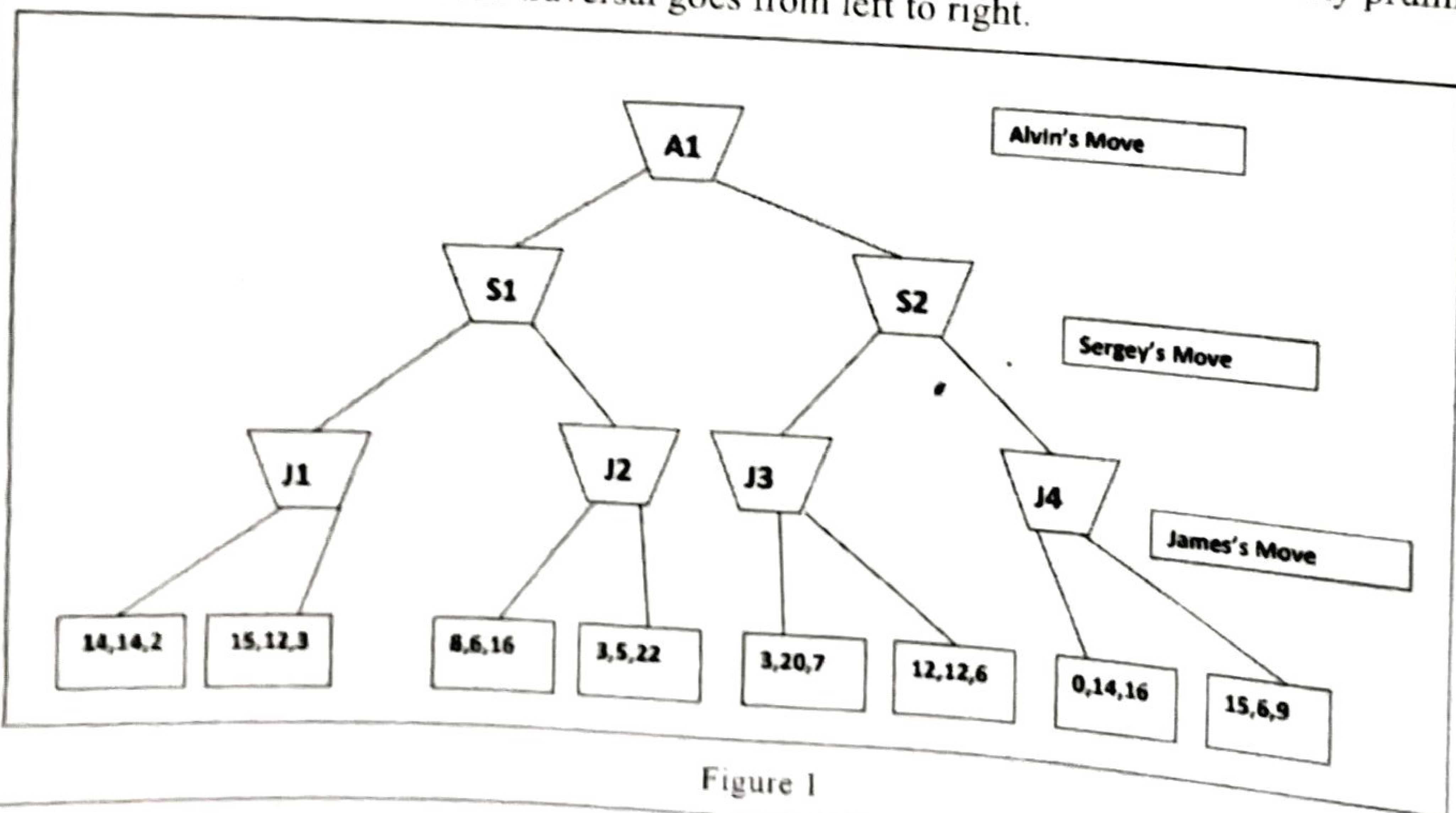


Figure 1



**Q2 Write the answer of the following questions**  
 The following is a dataset of patients who have tested positive or negative for COVID-19. Their symptoms are cough (C), fever (F), difficulty breathing (B), and tiredness (T).

|           | Cough (C) | Fever (F) | Difficulty breathing (B) | Tiredness (T) | Diagnosis |
|-----------|-----------|-----------|--------------------------|---------------|-----------|
| Patient 1 |           | X         | X                        | X             | Sick      |
| Patient 2 | X         | X         |                          | X             | Sick      |
| Patient 3 | X         |           | X                        | X             | Sick      |
| Patient 4 | X         | X         |                          | X             | Healthy   |
| Patient 5 | X         |           | X                        |               | Healthy   |
| Patient 6 |           | X         |                          |               | Healthy   |
| Patient 7 |           | X         |                          | X             | Healthy   |
| Patient 8 |           |           |                          |               |           |

The goal of this question is to build a Naive Bayes model that predicts the diagnosis from the symptoms. Use the Naive Bayes algorithm to find the following probabilities:

- The probability that a patient is sick given that the patient has a cough
- The probability that a patient is sick given that the patient is not tired
- The probability that a patient is sick given that the patient has a cough and a fever
- The probability that a patient is sick given that the patient has a cough and a fever, but no difficulty breathing

**NOTE:** 1. Follow the steps of Naïve Bayes completely, don't try to solve it by shortcut method. 'X' means Yes

- For the above questions, the symptoms that are not mentioned are completely unknown to us. For example, if we know that the patient has a cough, but nothing is said about their fever, it does not mean the patient doesn't have a fever.

**Q3 Write the answer of the following question**

Consider the graph shown in the Figure 2. We can search it with a variety of different algorithms, resulting in different search trees. Each of the trees (labelled G1 through G7) was generated by searching this graph, but with a different algorithm. Assume that children of a node are visited in alphabetical order. Each tree shows all the nodes that have been visited. Numbers next to nodes indicate the relevant "score" used by the algorithm for those nodes. For each tree, indicate whether it was generated with

- Depth first search
- Breadth first search
- Uniform cost search
- A\* search
- Best-first (greedy) search

In all cases a strict expanded list (open list i.e. the node that is not yet explored) was used. Furthermore, if you choose an algorithm that uses a heuristic function, say whether we used

H1: heuristic 1 = {h(A) = 3, h(B) = 6, h(C) = 4, h(D) = 3}

H2: heuristic 2 = {h(A) = 3, h(B) = 3, h(C) = 0, h(D) = 2}

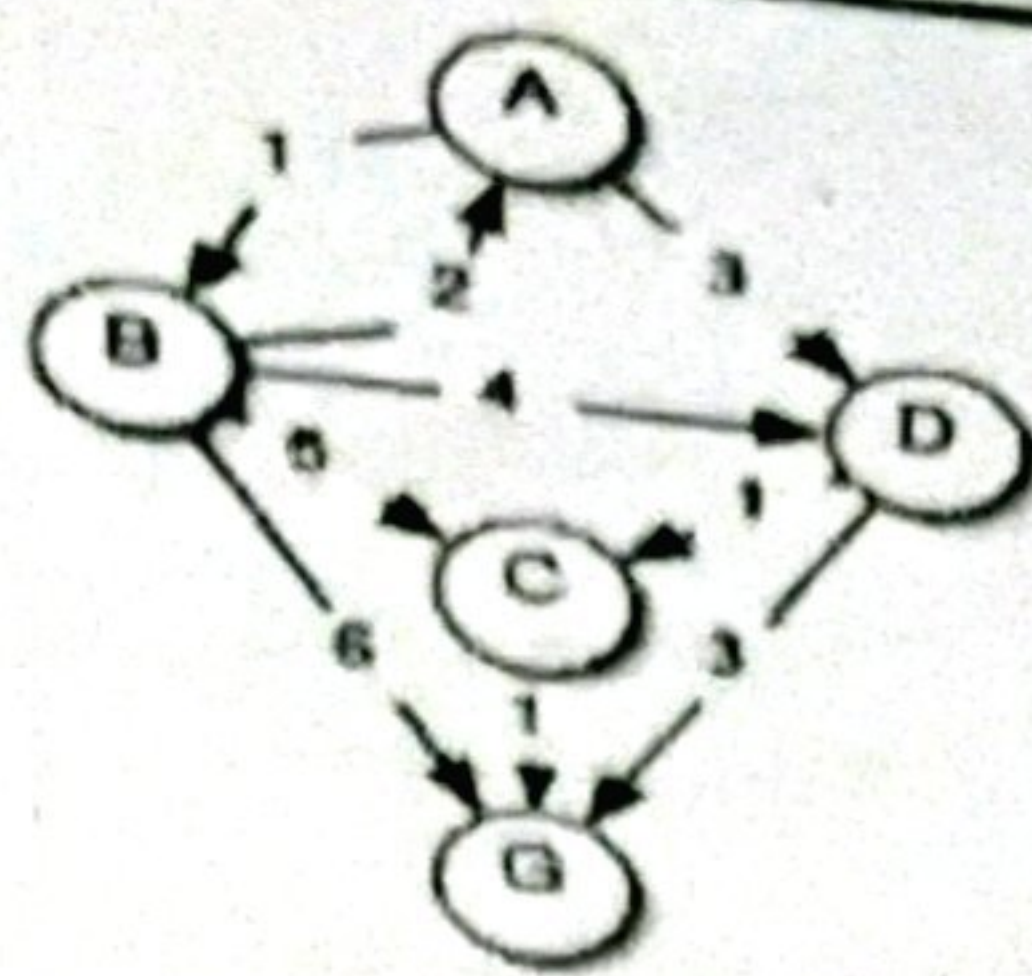
**Write your answers** for all trees on the basis of following parameters:

Tree Number (e.g. G1):

- Algorithm Name:
- Heuristic (if any):
- Did it find least-cost path? If not, why? :

Also, for all algorithms, say whether the result was an **optimal path** (measured by sum of link costs), and if not, why not. Be specific.





|   | H1 | H2 |
|---|----|----|
| A | 3  | 3  |
| B | 6  | 3  |
| C | 4  | 0  |
| D | 3  | 2  |

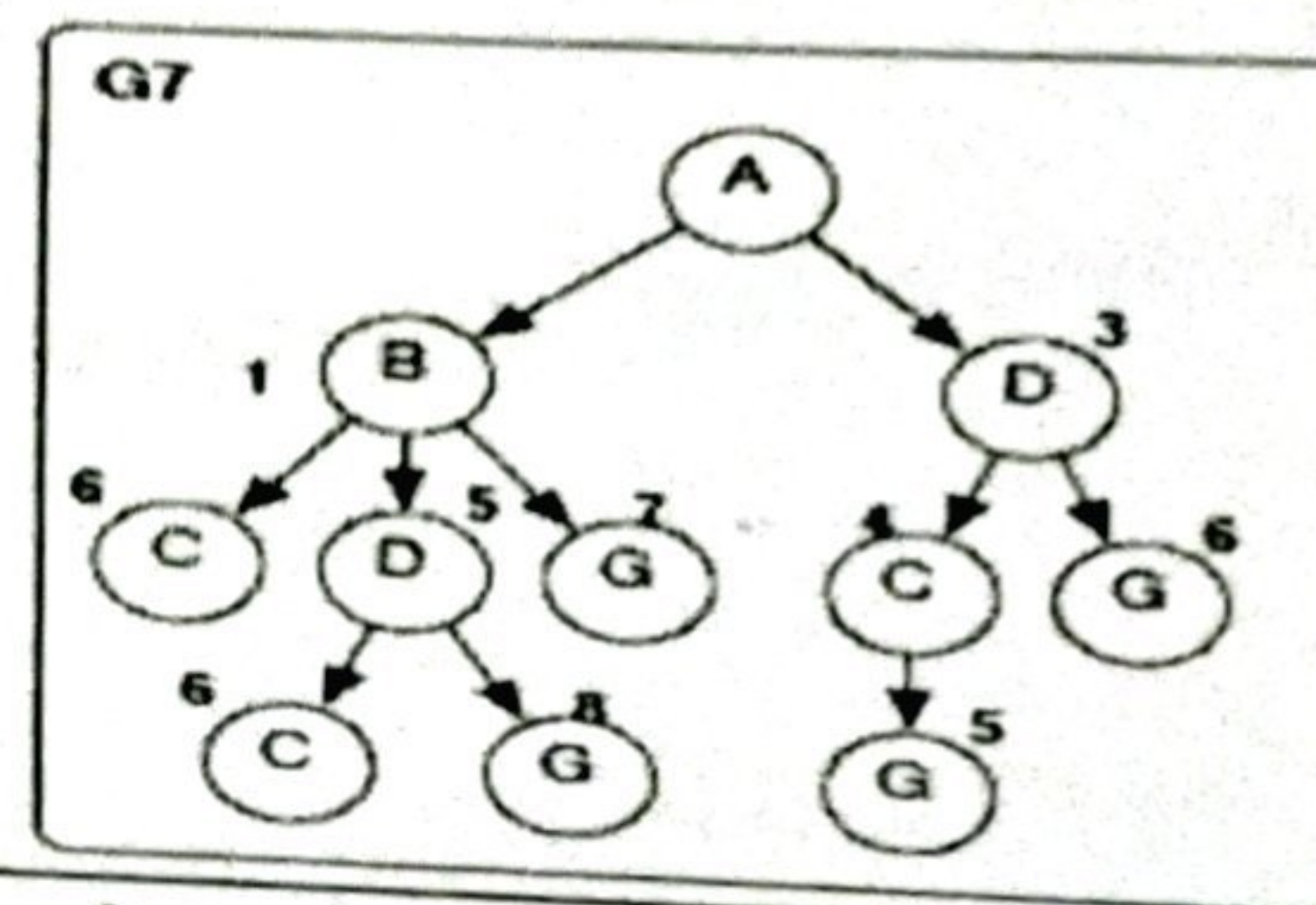
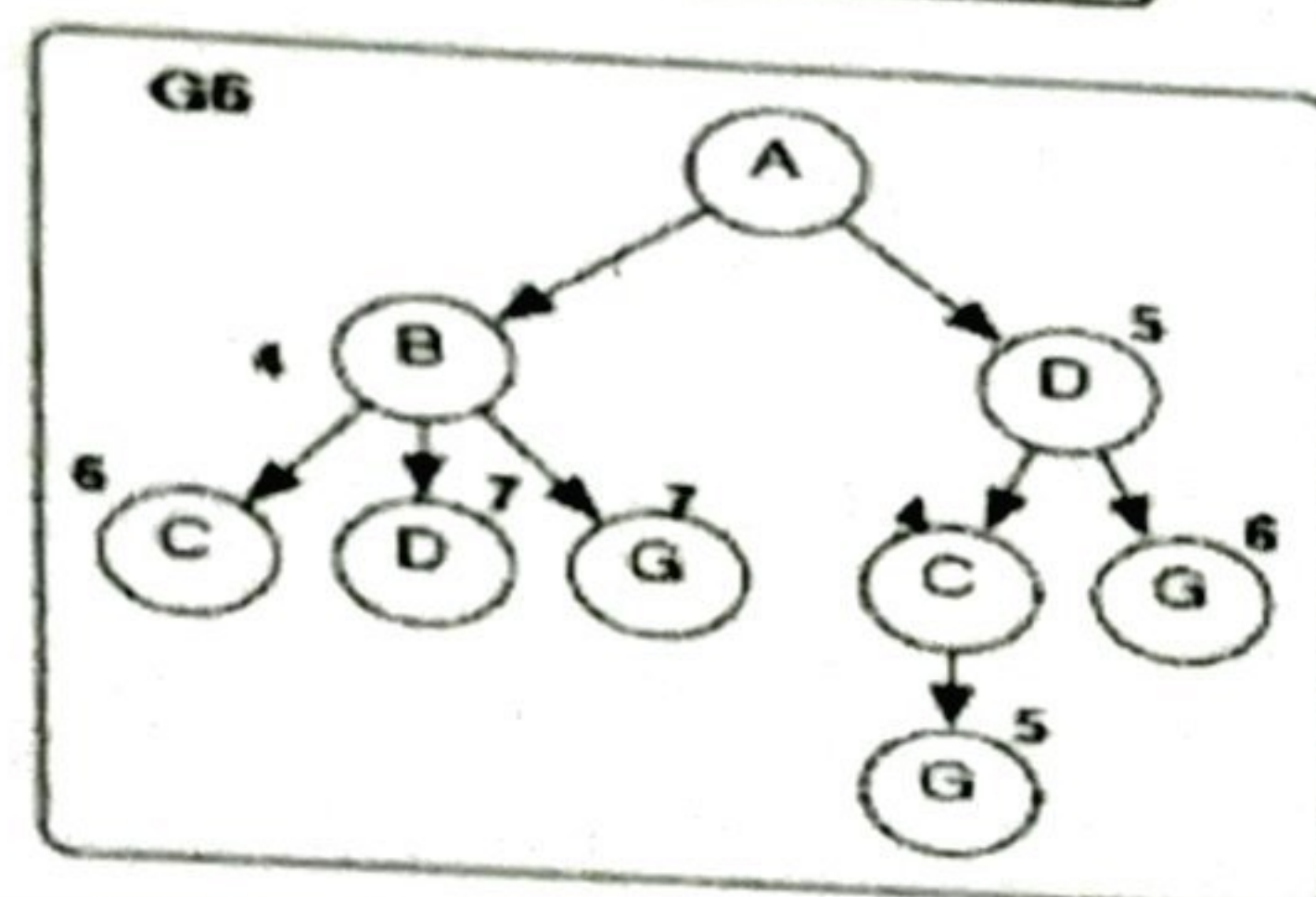
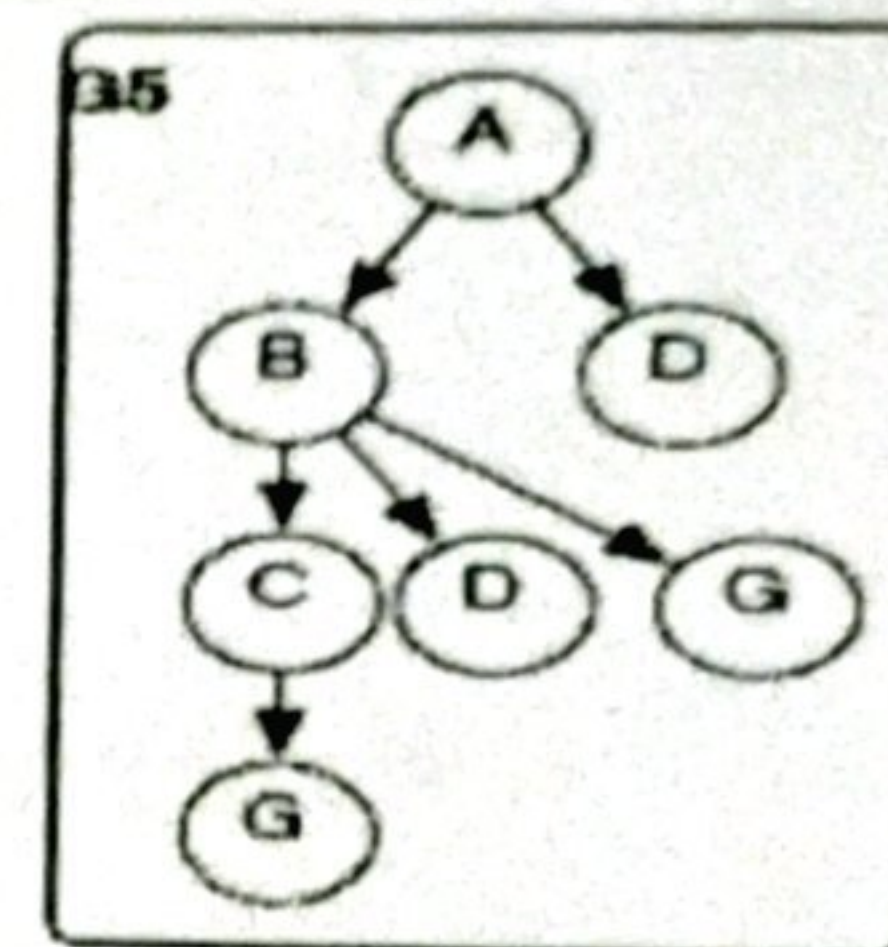
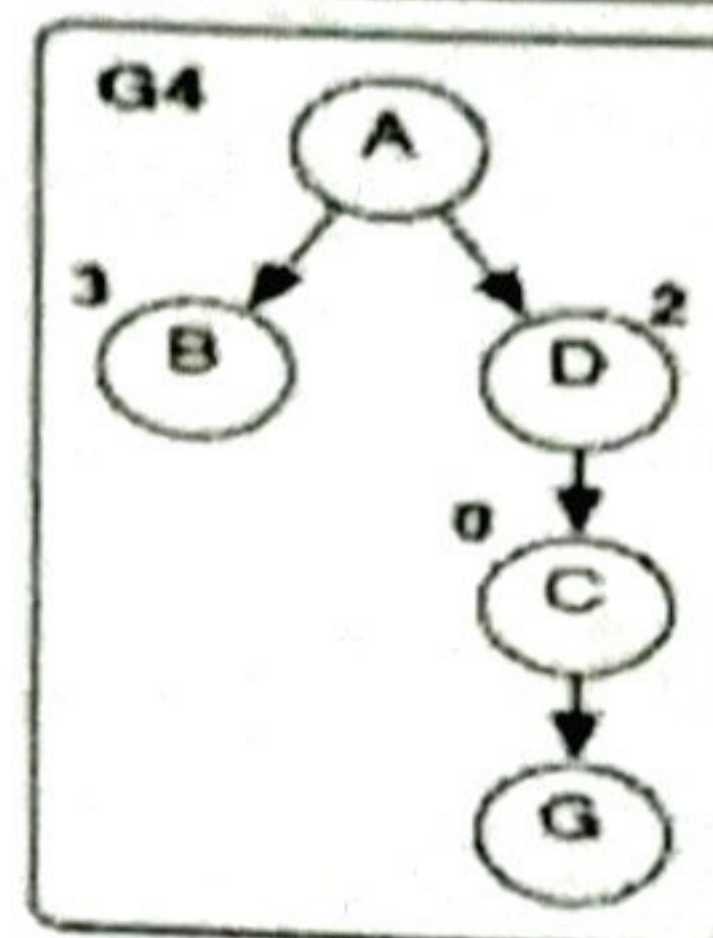
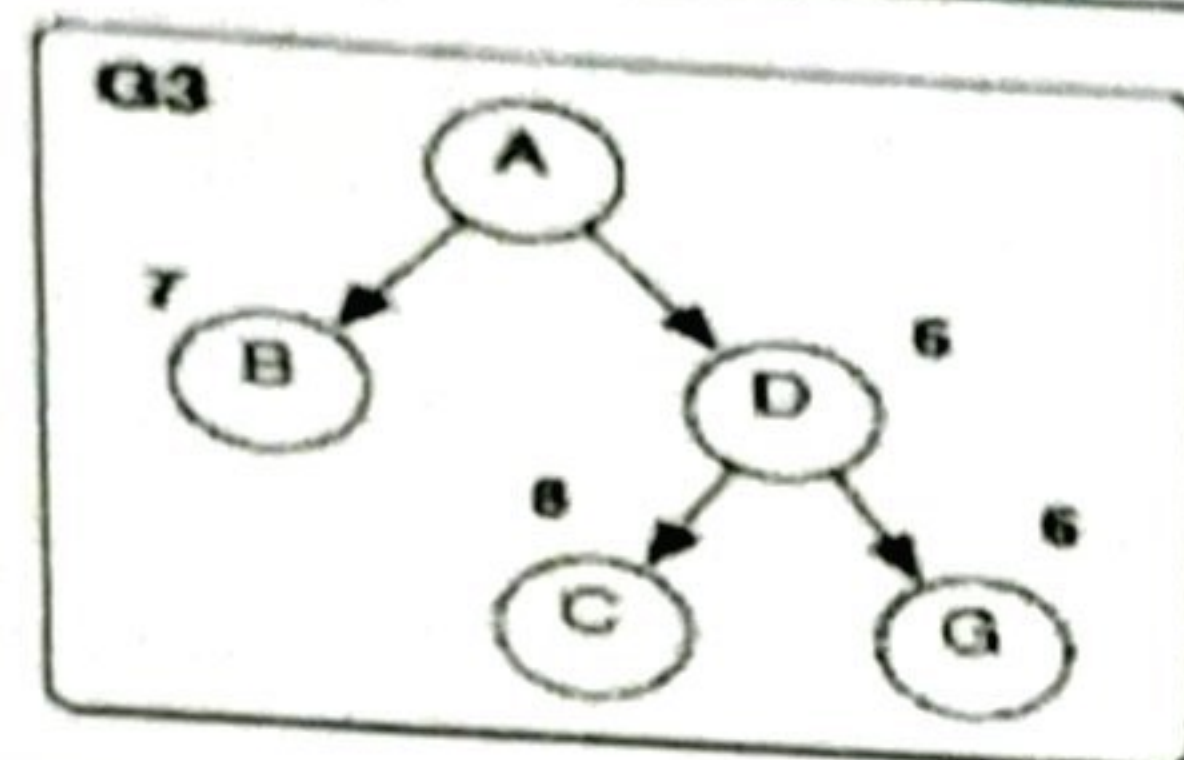
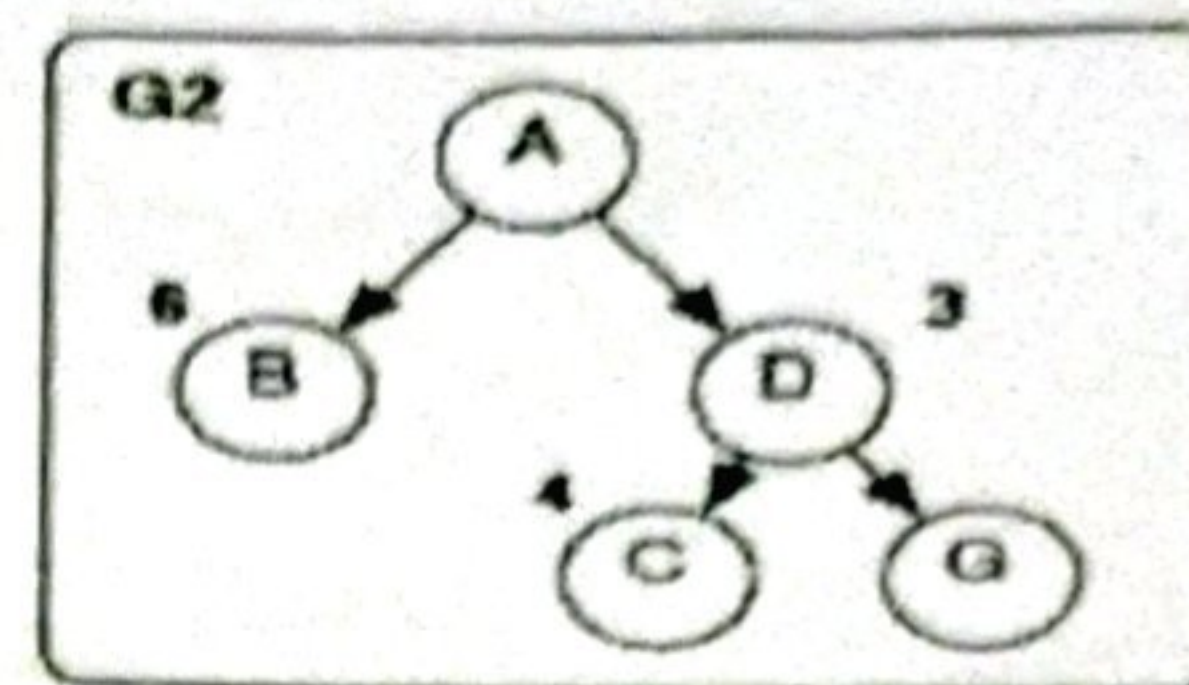
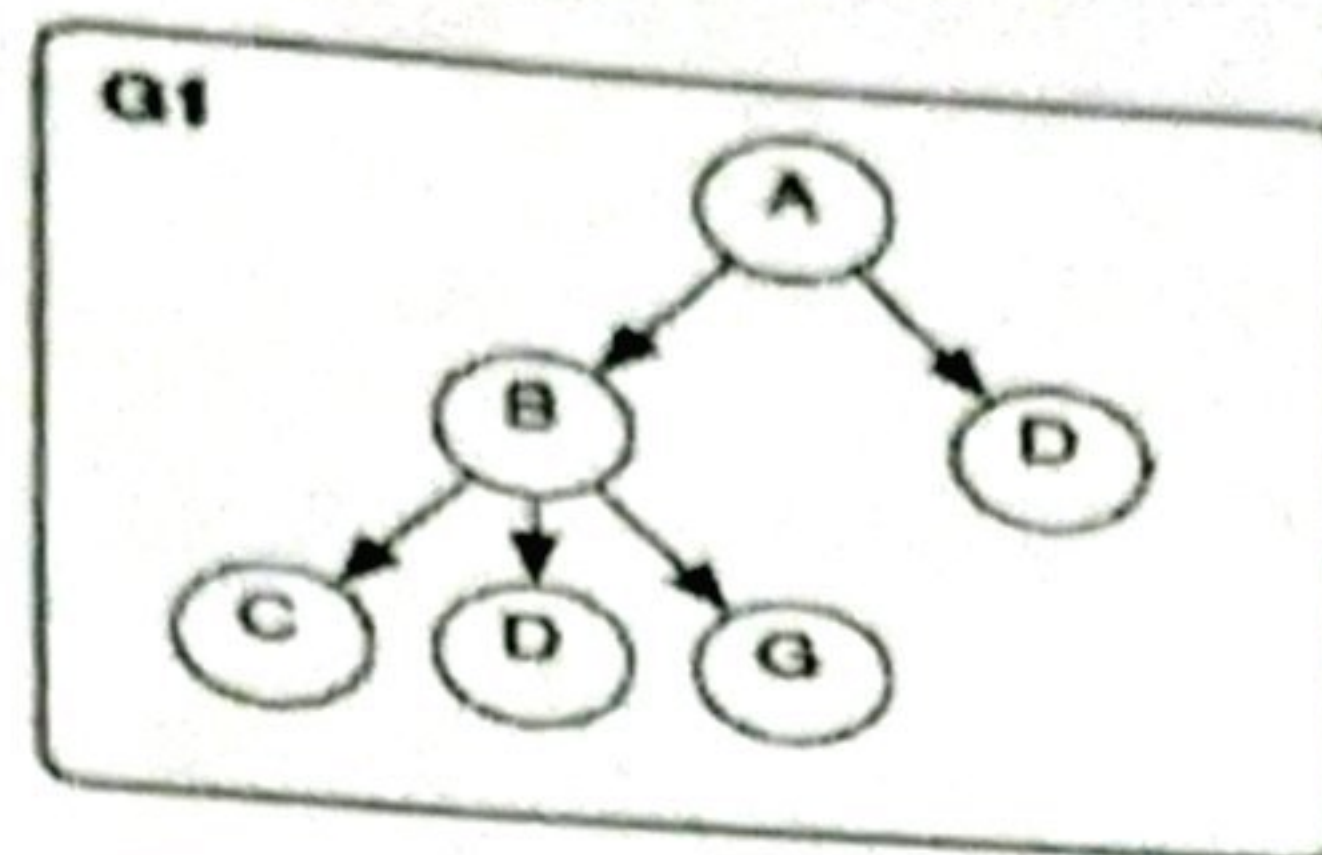


Figure 2

**Q4** Write the answer of the following questions

1. Imagine a scenario where a robot is navigating a maze with dynamically shifting walls. How would you design an effective heuristic function for the A\* search algorithm to guide the robot towards the goal state when the true goal position is unknown and subject to frequent changes?
2. How does the selection of appropriate knowledge representation techniques in artificial intelligence systems impact the efficiency and effectiveness of problem-solving tasks, and what are different types of knowledge representation?

---ALL THE BEST---





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Department of Computer Science & Engineering  
Mid Semester Examination, Session 2023-24 (Even)

Semester: IV

Programme: B.Tech.

Branch: CSE

Course Name: Micro Processor and Its Application

Course Code: CSN 14400

Max. Marks: 25

Time: 1.5 Hours

Registration No.:

|   |   |   |   |  |  |  |  |  |  |
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| 2 | 0 | 2 | 2 |  |  |  |  |  |  |
|---|---|---|---|--|--|--|--|--|--|

Instructions:

1. Attempt all five questions. Each question carries equal mark.
2. All parts of a question should be answered in one attempt SEQUENTIALLY.

- |    |                                                                                                                                                                                                                                                                                                            | Marks          |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Q1 | a. Write a tabular representation of 8085 machine cycle, status, and control signals                                                                                                                                                                                                                       | 3+2<br>[CO1]   |
|    | b. Explain the following instructions with example<br>(i) DAA (ii) CMP                                                                                                                                                                                                                                     |                |
| Q2 | a. Draw and explain the 8085 timing diagram for execution of the instruction MVI A, 32H. Assume that the instruction is stored starting at the memory location 2000H. Also assume the Hex Code for opcode is 3EH.                                                                                          | 5<br>[CO2]     |
| Q3 | a. Explain the memory address range of 4K (4096 × 8) memory chip. The size of address bus is 16 bits.                                                                                                                                                                                                      | 1+3+1<br>[CO2] |
|    | b. Suppose the 4K bytes of memory are available in 4 chips of 1K byte each. Calculate the address range for all the chips. How many such chips can be addressed using 16 bit address lines? Specify their address range..                                                                                  |                |
|    | c. The memory address of the last location of a 1K byte memory chip is given as FBFFH. Specify the starting address. The size of address bus is 16 bits.                                                                                                                                                   |                |
| Q4 | a. Write an assembly language program to count continuously in hexadecimal from 00H to FFH in a system with a 1 μs clock period. Use 8-bit registers to set up a delay of 2ms between each count and display output at one of the output ports.                                                            | 5<br>[CO3]     |
| Q5 | a. Write a program in assembly level language where 10 numbers are stored starting at memory location 4000H. Find the cumulative sum of all the positive numbers. Store the result at memory location 5000H followed by the number of positive elements added. Also store the number of negative elements. | 5<br>[CO3]     |